

A New Analysis of the Morphological Structure of Japanese Verbal Adjectives

Shinya Makino

itakanro@g.ecc.u-tokyo.ac.jp

Abstract:

The present paper reanalyses the morphological structure of Japanese verbal adjectives (VAs) — simply adjectives, or *i*-adjectives. Traditionally, inflected forms of verbal adjectives, such as *takaku* — so-called *ren'yōkei* — from the VA *takai* "to be high/tall", have been analysed as consisting of the so-called stem *taka-* and the so-called ending *-ku*. However, this analysis appears to have limitations. Considering **the morphological, syntactic, morpho-syntactic and semantic parallelism with verb stems** — such as *tabe* from the verb *taberu* "to eat" — it can be inferred that ***takaku* as a whole corresponds to the stem of the verbal adjective**. This paper thus analyses *taka-* as one of the two immediate constituents of the VA stem *takaku*, which denotes, as a VA radical, the core lexical meaning of the verbal adjective and *-ku* as the other constituent, functioning as the VA thematic suffix — namely, VA stem-forming suffix — that licences the VA radical as a syntactic constituent within the VA category at the sentential level. The paper then examines how verbal adjectives encode the tense value [–PAST], proposing a new interpretation based on historical and morphological perspectives. Rather than the traditional interpretation that *-i*, as in *takai*, expresses non-past, it is argued that the phonologically zero inflexional suffix *-0*, as in [[[*taka*]*i*]]*0*, denotes [–PAST] through opposition to *-ta*, specified as [+PAST], as in [[[*taka*]*k*]*at*]*ta*. Based on this, the paper further explores the allomorphy of the VA thematic suffix *-i* $\approx$ *-(k)i* $\approx$ *-ku* $\approx$ *-ke* and investigates how the distribution of these allomorphs is determined by their syntactic environments. Finally, the paper explores the similarities between the morphological structures of verbal adjectives and true verbs — particularly vowel-stem verbs.

1. Introduction

The inflected forms of Japanese verbal adjectives — simply adjectives, or *i*-adjectives — are not so transparent in their morphological structure as those of verbs. In verbs, the boundary between the stem and the ending can be clearly determined. Taking the verb *taberu* "to eat" as an example, the non-past form ***taberu*** consists of the stem *tabe-* and the inflexional ending composed of the suffix *-ru*, specified as [–PAST], while the past form ***tabeta*** consists of the same stem *tabe-* and the inflexional ending composed of the suffix *-ta*, specified as [+PAST]. In other words, the opposition between the two forms is reduced to a morphologically transparent opposition between the suffixes *-ru* and *-ta*, both sharing the same stem *tabe-* as their inflexional base.

With verbal adjectives, on the other hand, taking *takai* "to be high/tall" as an instance, the non-past form is ***takai***, while the past form is ***takakatta***. Their relationship is not so simple as the opposition between the non-past and past forms of verbs, and the boundary between the stem and the ending is difficult to define. Tsujimura (2007: 119) states: "adjectives are identified by a variety of conjugational endings, such as the non-past ending *-i*,... and the past tense ending *-kat-ta*". Following this, in her analysis of the VA *takai*, she treats the non-past form *takai* as the stem *taka* + the ending *-i*, and the past form *takakatta* as the stem *taka-* + the ending *-kat-ta*. This seems to be the traditional and standard analysis in Japanese linguistics.

This paper, in contrast, proposes a different type of analysis, assuming that while the formation of the basic stem of verbal adjectives — the VA stem — takes place at the level of the lexicon, the combination of that basic stem and the inflexional ending occurs at the level of syntax, because such endings are assumed to constitute the phrasal head within the sentence structure. Based on this assumption, the morphological structure is analysed as in (1), where –, () and + indicate suffixation, deletion and compounding, respectively:

(1)

Stem: sentential			Ending
VA stem: lexical		Light verb stem	Inflexional suffix(es)
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-i</i>	—	<i>-0</i> [–PAST]
<i>taka-</i>	<i>-ku</i>	—	— [0 PAST]
<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-ta</i> [+PAST]
<i>taka-</i>	<i>-ke</i>	—	<i>-reba</i>

As mentioned above, the conclusive form *taka-i* — so-called *syūsikei* in Japanese — is traditionally analysed as the stem *taka-* + the ending *-i*, but here, *taka-i* as a whole is treated as the lexical-level stem — the VA stem. The root *taka-* constitutes, then, the VA radical and *-i* is analysed as the VA thematic suffix — namely, the VA stem-forming suffix. To this VA stem *taka-i*, the phonologically zero inflexional ending *-0*, specified as [–PAST], is attached.

Similarly, the conjunctive or adverbial form *taka-ku* — so-called *ren'yōkei* — is generally analysed as the stem *taka-* + the ending *-ku*, but here, *taka-ku* as a whole is likewise treated as the

VA stem, with *taka-* being the VA radical and *-ku* the VA thematic suffix. Unlike *taka-i-0*, however, *taka-ku* does not have any ending and is analysed as consisting solely of the stem, due to its atemporality, much like the conjunctive form *tabe* from the verb *tabe-ru*.

On the other hand, in *taka-k+at-ta* — which is generally analysed as the stem *taka-* + the complex ending *-kat-ta* — *taka-k* from *taka-ku* is considered the lexical-level VP stem. When the VA stem *taka-ku* combines with the tense suffix *-ta* specified as [+PAST] at the sentential level, the stem *ar-* from the light verb *ar-u* is inserted immediately after it, due to the *taka-ku*'s incompatibility with the verbal suffix *-ta*. This results in the final form *taka-k+at-ta*, with the deletion of *-u* in *taka-ku* before *ar-* — which indicates the compounding of *taka-ku* with *ar-* — and the phonological change *ar-/ar-/* → *at-* [at-].

Finally, the conditional form *taka-ke-reba* — so-called *kateikei* — is generally analysed as the stem *taka-* + the ending *-kereba*. Here, however, *taka-ke* as a whole is treated as the VA stem, with *taka-* being the VA radical and *-ke* the VA thematic suffix, while *-reba* is exactly the same inflexional suffix found in the verbal conditional form *tabe-reba* from *tabe-ru*. The following will examine the justification for this analysis presented in (1).

2. The identification of the stem of verbal adjectives

Traditionally, as has been repeated, the conclusive form of verbal adjectives has been analysed as consisting of a stem followed by the inflexional ending *-i* — e.g. *taka-i* consisting of *taka-* + *-i*. On the other hand, the conclusive form of verbs — particularly vowel-stem verbs — has been analysed as consisting of a stem followed by the inflexional ending *-ru*. In *tabe-ru*, for instance, *tabe-* is considered the stem and *-ru* the ending. **Nevertheless, although *taka-* and *tabe-* have been both equally treated as stems, their syntactic behaviour differs significantly.** The verbal stem *tabe-* without overt ending can appear on its own within a sentence as in (2), whereas the so-called adjectival stem *taka-* can never occur independently in such contexts as shown in (3):

(2) *Kare wa gohan o tabe, (sosite) deteita / deteiku.* "He, having a meal, (then) left / leaves."

(3) **Kanozyo wa se ga taka, (sosite) utukusikatta / utukusii.* (ungrammatical)

"She, being tall, was / is (also) beautiful."

What can occur in the same syntactic context as the verbal stem *tabe-* is not the so-called adjectival stem *taka-* itself, but rather the conjunctive form *taka-ku*, as in (4), which has traditionally been analysed — like *taka-i* — as consisting of the so-called stem *taka-* and the so-called inflexional ending *-ku*:

(4) *Kanozyo wa se ga taka-ku, (sosite) utukusikatta / utukusii.* (grammatical)

"She, being tall, was / is (also) beautiful."

Accordingly, the verbal *tabe* and the adjectival *taka-ku* share the same syntactic property of being **able to occur at the end of non sentence-final coordinated clauses**. Moreover, as is clear from (2) and (4), at the stage of *Kare wa gohan o tabe* and *Kanozyo wa se ga taka-ku* in each syntactic string, it remains indeterminate whether the utterance refers to a past or non-past situation. Temporal specification is only achieved by the final predicate — *deteitta* / *utukusikatta* in the past and *deteiku* / *utukusii* in the non-past. In this respect as well, the verbal stem *tabe-* and the complex form *taka-ku* share the same semantic property of being **atemporal**: they do not themselves make any temporal reference. Furthermore, the similarity between the verbal stem *tabe-* and the complex *taka-ku* is not limited to these points; as shown in (5) - (10), both **can appear in the sentence followed by the inflexional suffixes -te, -temo and -tatte**.

- (5) *Kare wa gohan o tabe-te, deteitta / deteiku*. "He, having a meal, then left / leaves."
- (6) *Kanozyo wa se ga taka-ku-te, utukusikatta / utukusii*. "She, being tall, was / is beautiful."
- (7) *Kare wa gohan o tabe-temo, hutoranai*. "He, even having a meal, does not gain weight."
- (8) *Kanozyo wa se ga taka-ku-temo, medatanai*. "She, even being tall, does not stand out."
- (9) *Kare wa gohan o tabe-tatte, hutoranai*. "He, even if he eats, does not gain weight."
- (10) *Kanozyo wa se ga taka-ku-tatte, medatanai*. "She, even if tall, does not stand out."

In contrast, the forms **taka-te*, **taka-temo* and **taka-tatte*, each composed of the so-called stem *taka-* followed by an inflexional suffix, are all ungrammatical. In this way, **there is clear syntactic, semantic and morpho-syntactic parallelism between *tabe-* and *taka-ku***. Given that *tabe-* is analysed as the stem of the verb, it seems more appropriate to treat *taka-ku* — not as the combination of the so-called stem *taka-* and the so-called inflexional ending *-ku*, as traditionally assumed — but rather **as the stem of the verbal adjective in its entirety**, which we will henceforth refer to as **the VA stem**. This raises the following question: how, then, should *taka-* — traditionally regarded as the stem — and *-ku* — traditionally regarded as the inflexional ending — be reanalysed morphologically as inflexional constituents? This issue will be discussed in the following section.

3. The radical and the thematic suffix of verbal adjectives

When we observe the inflected forms of the verbal adjective *taka-i* — namely, *taka-i* $\approx$ *taka-(k)i* $\approx$ *taka-ku* $\approx$ *taka-ku-te* $\approx$ *taka-ku-temo* $\approx$ *taka-ku-tatte* $\approx$ *taka-k+at-ta* $\approx$ *taka-k+at-tar-ō* $\approx$ *taka-k+at-tar-a(ba)* $\approx$ *taka-k+at-tar-i* $\approx$ *taka-k+ar-ō* $\approx$ *taka-k+ar-e* $\approx$ *taka-ke-reba* — we find that they all share the element *taka-*. This *taka-* is also attested in derived nouns such as *taka-sa* "height" and *taka-mi* "loftiness" as well as in compound nouns such as *taka+nozomi* "high hope" and *taka+dai* "high platform" (- and + indicate suffixation and compoundig, respectively), where

it refers to the very property or state of "being high/tall". Since it cannot be further decomposed into smaller morphemes, it is to be regarded as a root. However, as the ungrammaticality of (11) clearly shows, the root *taka-* alone cannot function as a verbal adjective — that is, it cannot be inserted into a terminal node dominated by the VA node — and thus cannot establish structural relations with other constituents within the sentence:

(11) **Koko kara, taka sobieru tō ga mieru.* (ungrammatical)

"From here, a tower rising high can be seen."

By contrast, example (12), in which *taka-* is accompanied by *-ku*, is grammatical:

(12) *Koko kara, taka-ku sobieru tō ga mieru.* (grammatical)

"From here, a tower rising high can be seen."

Now, as the derived words and compound words cited above show, **the root *taka-* functions only as a mere constituent of word formation**: it cannot itself serve as a syntactic unit at the sentence level. We are thus led to posit that *taka-* requires an additional element in order to function as a syntactic category VA at the sentence level. This element is precisely *-ku*, which is present in the sentential form *taka-ku*, but absent from word-level forms such as *taka-sa*, *taka-mi*, *taka-nozomi* and *taka-dai*. As observed in Section 2, the addition of *-ku* to the root *taka-* yields a verbal adjective stem that is functionally — more precisely, syntactically, semantically and morpho-syntactically — equivalent to the verbal stem *tabe-*. We shall therefore refer to *-ku* as the **VA thematic suffix** — namely, the **VA stem-forming suffix**.

The historical development of verbal adjectives also supports this interpretation: "Originally, adjectives, remaining as stems, were positioned before nouns and verbs, modifying them, as in *nagauta*, *takayuku*. However, to explicitly express the functions of the conjunctive, attributive and conclusive forms, inflected forms such as *takaku*, *takaki*, *takasi* were established. Moreover, after the Nara period, forms such as *takakere* developed as realis form *izenkei*. In this way, adjectives performing the role of predicates developed more slowly" (Ōno, S., et al. (Eds.). 1990: 1467, translated from Japanese). From the perspective of this paper, the VA radical — e.g. *naga-* "long" and *taka-* "high/tall" — originally functioned only as a formative element at the lexicon level, forming compounds such as *naga+uta* "long poem" and *taka+yuku* "to fly high into the sky", placed before base nouns or base verbs. However, in order to function as the head of a VA phrase at the sentential level, that is, to function as a category VA, the thematic suffixes *-ku*, *-ki*, *-si* — corresponding to *-ku*, *-(k)i*, *-i* in Contemporary Japanese — were added.

On the other hand, *taka-* was previously referred to as a root in relation to *-ku*, but this is not strictly accurate from the perspective of morphological structure. As *na-daka-* — derived from

na "name" + *taka-* "high", with *rendaku* in the second element — in the compound verbal adjective *na-daka-i* "to be famous" in (13) shows, what precedes the VA thematic suffix need not be a single root, but may also involve root-root compounding:

(13) *Yamazaki wa **na-daka-i** genggokusya da.* "Yamazaki is a famous linguist."

It is therefore morphologically inappropriate to refer to the constituent preceding the VA thematic suffix *-ku* as a root. In what follows, we shall refer to it as a **VA radical**, in the sense of "one of the two immediate constituents of a verbal adjective stem that constitutes its semantic core".

As the inflected forms *taka-ku*, *taka-ku-te*, *taka-ku-temo* and *taka-ku-tatte* show, the verbal adjective stem formed with the suffix *-ku* — namely, *taka-ku* — can appear on its own, or followed by the suffixes *-te*, *-temo* and *-tatte*. From this, the morphological structure of verbal adjectives may be analysed as in (14):

(14)

VA stem		Ending
VA radical	VA thematic suffix	Inflexional suffix
<i>taka-</i>	<i>-ku</i>	—
<i>taka-</i>	<i>-ku</i>	<i>-te</i>
<i>taka-</i>	<i>-ku</i>	<i>-temo</i>
<i>taka-</i>	<i>-ku</i>	<i>-tatte</i>

It can be confirmed that the morphemes *-te*, *-temo*, *-tatte* are inflexional suffixes as follows. In *taka-ku-te*, *taka-ku-temo*, *taka-ku-tatte*, it is possible to insert syntactically the focus particle *-mo* "too, also" or the topicalisation/contrast particle *-wa* "as for, etc." immediately after the verbal adjective stem *taka-ku-*. In these cases, the stem *at-* from the light verb *ar-u* "to be" is always inserted before *-te*, *-temo*, *-tatte*, as in (15), where # denotes a word boundary:

(15) *taka-ku-te* → *taka-ku#mo#at-te* vs. **taka-ku#mo#te* "being also high/tall"
taka-ku-temo → *taka-ku#wa#at-temo* vs. **taka-ku#wa#temo* "high/tall, though it is so"
taka-ku-tatte → *taka-ku#wa#at-tatte* vs. **taka-ku#wa#tatte* "high/tall, even if it is so"

In the case of the verbal forms *tabe-te*, *tabe-temo*, *tabe-tatte* in (5), (7), (9), the situation is exactly the same, as shown in (16):

(16) *tabe-te* → *tabe#mo#si-te* vs. **tabe#mo#te* "doing also eat"
tabe-temo → *tabe#wa#si-temo* vs. **tabe#wa#temo* "eat, though it does"
tabe-tatte → *tabe#wa#si-tatte* vs. **tabe#wa#tatte* "eat, even if it does"

In this case, the stem *si-* from the light verb *su-ru* "to do" is inserted before *-te*, *-temo*, *-tatte*. From

(15) and (16), the following observations (17)–(20) may be drawn:

(17) In *taka-ku-te*, *taka-ku-temo*, *taka-ku-tatte* in (15) and *tabe-te*, *tabe-temo*, *tabe-tatte* in (16), respectively, it is possible to insert the particles *mo* and *wa* immediately after the VA stem *taka-ku-* and the V stem *tabe-* at the syntactic level. This indicates that *taka-ku-* and *tabe-* occupy the same syntactic structural position and, here too, **syntactic parallelism is observed between *tabe-* and *taka-ku-***.

(18) In both (15) and (16), if *-te*, *-temo*, *-tatte* were free morphemes and could constitute autonomous words, then forms such as **taka-ku#mo#te*, etc. and **tabe#mo#te*, etc. should be grammatical as they are. This is not the case, however: only when the stems of the light verbs *at-* and *si-* are inserted, as in *taka-ku#mo#at-te*, etc. and *tabe#mo#si-te*, etc., do the forms become adequate. It follows, then, that *-te*, *-temo*, *-tatte* are inflexional suffixes which cannot constitute a word independently and, formally, always require a stem.

(19) Based on (18), in *taka-ku-te*, etc. in (15) and *tabe-te*, etc. in (16), the stems required by the inflexional suffixes *-te*, *-temo*, *-tatte* are *taka-ku-* and *tabe-*, respectively. On the other hand, when the particles *mo* or *wa* are inserted, the inflexional suffixes *-te*, *-temo*, *-tatte* are separated from these stems, and since *mo* and *wa* cannot function as stems for these suffixes, the formally required stems *at-* and *si-* of the light verbs are inserted, as they can combine with the suffixes to form a word. Based on this, the relation *tabe- : si- = taka-ku- : at-* is established, and *taka-ku-*, like *tabe-*, is thereby judged to qualify as a stem; and, here too, **morphological parallelism is observed between *tabe-* and *taka-ku-***.

(20) In (15), the reason why it is the verb stem *at-* from the light verb *ar-u* that is inserted in *taka-ku#mo#at-te*, etc. is that Japanese lacks a class of light verbal adjectives and their corresponding stems, and hence a verbal stem must be employed to support formally the suffixes *-te*, *-temo*, *-tatte*. But why is it not the stem *si-* from *su-ru* that is selected — namely, *taka-ku#mo#si-te*, etc. — but rather the stem *at-* from *ar-u* — namely, *taka-ku#mo#at-te*, etc.? The motivation seems to be that the prototypical categorical distinction — stativity versus dynamicity — between the verbal adjective stem and the verb stem is reflected in the opposition between the stative verb *ar-u* "to be" and the dynamic verb *su-ru* "to do".

Together with the description at the end of Section 2, it can be observed that **between the verbal stem *tabe-* and the verbal adjective stem *takaku-*, clear and multiple parallelism — morphological, syntactic, morpho-syntactic and semantic — is evident.**

Next, let us turn to more complex structures: in the inflected forms *taka-k+at-ta*, *taka-k+at-tar-ō*, *taka-k+at-tar-a(ba)*, *taka-k+at-tar-i*, *taka-k+ar-ō*, *taka-k+ar-e*, if the focus particle *mo* "too, also" is inserted before the conjugated forms of the light verb *ar-u* "to be" — namely, *at-ta*, *at-tar-ō*, *at-tar-a(ba)*, *at-tar-i*, *ar-ō* and *ar-e* — and the string is thereby divided into two words, the

following alternations in (21) can be observed:

(21) <i>taka-k+at-ta</i> $\approx$ <i>taka-ku#mo#at-ta</i>	"It was high/tall" $\approx$ "... also high/tall"
<i>taka-k+at-tar-ō</i> $\approx$ <i>taka-ku#mo#at-tar-ō</i>	"It would be high/tall" $\approx$ "... also high/tall"
<i>taka-k+at-tar-a(ba)</i> $\approx$ <i>taka-ku#mo#at-tar-a(ba)</i>	"If it is/were high/tall" $\approx$ "... also high/tall"
<i>taka-k+at-tar-i</i> $\approx$ <i>taka-ku#mo#at-tar-i</i>	"Sometimes it is high/tall" $\approx$ "... also high/tall"
<i>taka-k+ar-ō</i> $\approx$ <i>taka-ku#mo#ar-ō</i>	"It will be high/tall" $\approx$ "... also high/tall"
<i>taka-k+ar-e</i> $\approx$ <i>taka-ku#mo#ar-e</i>	"Be high/tall" $\approx$ "... also high/tall"

From this, it can be seen that the VA thematic suffix *-ku* undergoes vowel deletion and surfaces as *-k* when it precedes the vowel-initial stem of the light verb *ar-u*, namely *ar-* $\approx$ *at-*. This vowel deletion can be taken to indicate that the VA stem and the light verb stem — each of which can function as an autonomous word — have come to form a single compound stem.

The reason why the inflected forms in (21) contain *ar-* or *at-* — the stem of the light verb *ar-u* — is that the endings *-ta*, *-tar-ō*, *-tar-a(ba)*, *-tar-i*, *-ō*, *-e*, which express tense or/and modality, combine only with purely verbal stems. In other words, *taka-ku* is a non-verbal stem and thus cannot combine directly with these endings. Consequently, the VA stem *taka-ku* combines with the stem *ar-* of the light verb *ar-u*, forming a compound verbal stem *taka-k+ar-* ($\leftarrow$ *taka-ku* + *ar-*) at the sentence level, which as a whole comes to possess verbal status. This is made possible by the fact that Japanese is a head-final language, whereby the light verb stem *ar-* determines the syntactic category of the entire compound *taka-k+ar-*. Under this interpretation, the structure of (14) may be more accurately represented as in (22), where -, () and + indicate suffixation, deletion and compounding, respectively, as in (1):

(22)	Stem: sentential		Ending
	VA stem: lexical		Inflexional suffix(es)
	VA radical	VA thematic suffix	
	<i>taka-</i>	<i>-ku</i>	—
	<i>taka-</i>	<i>-ku</i>	<i>-te</i>
	<i>taka-</i>	<i>-ku</i>	<i>-temo</i>
	<i>taka-</i>	<i>-ku</i>	<i>-tatte</i>
	<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>
	<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>
	<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>
	<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>
	<i>taka-</i>	<i>-k(u)</i>	<i>+ar-</i>
	<i>taka-</i>	<i>-k(u)</i>	<i>+ar-</i>

Besides the both morphologically and phonologically conditioned allomorphy *-ku* $\approx$ *-k*, the VA thematic suffix exhibits another type of allomorphy *-ku* $\approx$ *-i* $\approx$ *-(k)i* $\approx$ *-ke* depending on the

syntactic environment in which it occurs, as shown in (23)–(28):

- (23) **Kanozyo wa se ga taka.*
(24) *Kanozyo wa se ga taka-i.* "She is tall."
(25) **Asoko ni se no taka otoko ga iru.*
(26) *Asoko ni se no taka-i otoko ga iru.* (Classical: *taka-ki*) "There is a tall man over there."
(27) **Se ga taka-reba ii to yū wake de wa nai.*
(28) *Se ga taka-ke-reba ii to yū wake de wa nai.* "It is not always the case that it is good if the height is tall."

Among these, *taka-ke* in (28), which bears the allomorph *-ke*, occurs in modern Japanese exclusively before the suffix *-reba*, which denotes a sequential conditional: hence, *taka-ke-reba* is referred to as the conditional form. By contrast, the syntactic conditions that determine the occurrence of *taka-i* in (24) and *taka-(k)i* in (26), bearing the allomorphs *-i* and *-(k)i* respectively, will be examined in the following section.

4. Syntactic conditions governing the allomorphy: $-i \approx -(k)i \approx -ku \approx -ke$

As observed in (4), the conjunctive form *taka-ku*, in which the VA thematic suffix appears in the form *-ku* and no inflexional ending follows, is atemporal. The fact that the form *taka-ku* lacks temporal reference is consistent with the interpretation that it does not bear any inflexional suffix expressing tense or modality. By contrast, the conclusive form *taka-i* and the attributive form *taka-(k)i* — *rentaikei* — appear to lack any inflexional ending just like the conjunctive form *taka-ku*, yet unlike *taka-ku*, they bear non-past value. The question, then, is how this non-past value is formally encoded. Traditionally, the final *-i* in *taka-i* and *taka-(k)i* has often been interpreted as the element that bears the temporal value of non-past. This interpretation is difficult to justify, however, either from a historical perspective or from the perspective of a descriptive analysis of modern Japanese.

Historically, verbal adjectives originally lacked a non-past vs. past tense opposition and, accordingly, they contained no formal element that encoded tense. For example, in Old Japanese of the 8th century, forms such as *taka-si*, *taka-ki* and *taka-ku* consisted, from the standpoint of the present analysis, solely of the stem, lacking any inflexional ending. However, probably due to the increasing need to mark "past" more explicitly, the past tense suffix *-ki* — which was solely compatible with verbal stems — and the stem *ar-* of the light verb *ar-i* were employed to constitute the past tense form *taka-k+ar-i-ki*. As Okimori (2017: 84) states, "The secondary conjugation known as *kari*-conjugation was formed by the conjunctive form being followed by the *rahen* verb *ari*, through the development *ku-ari* > *kari*. This made it possible for the form to attach to a wide

range of particles and auxiliary verbs" (translated from Japanese).

This suggests that the conclusive form *taka-si* and the attributive form *taka-ki*, which originally lacked any overt tense marker, came to express "non-past" through their formal opposition to the newly formed past tense *taka-k+ari-ki* — that is, by lacking the overt tense suffix *-ki*. In other words, from the perspective of formal opposition, the conclusive and attributive forms came to be analysed as bearing a zero inflexional suffix *-0* specified as [–PAST], which came to stand in opposition to the past-tense inflexional suffix *-ki* specified as [+PAST]. This is parallel to the case of English *cat-0*, where the absence of an overt plural marker in opposition to *cat-s* functions to denote the singular. Based on the above, the morphological structure of the classical forms *taka-si*, *taka-ki*, *taka-ku* and *taka-k+ar-i-ki* may be analysed as in (29):

(29)

Stem: sentential			Ending
VA stem: lexical		Light verb stem	Inflexional suffix
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-si</i>	—	-0 [–PAST]
<i>taka-</i>	<i>-ki</i>	—	-0 [–PAST]
<i>taka-</i>	<i>-ku</i>	—	— [0PAST]
<i>taka-</i>	<i>-k(u)</i>	+ar-i-	-ki [+PAST]

In the case of modern Japanese as well, it may be assumed that the partial system in (29) remains unchanged, with the past tense suffix *-ki* simply having been replaced by *-ta*, as shown in (30):

(30)

Stem: sentential			Ending
VA stem: lexical		Light verb stem	Inflexional suffix
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-i</i>	—	-0 [–PAST]
<i>taka-</i>	<i>-(k)i</i>	—	-0 [–PAST]
<i>taka-</i>	<i>-ku</i>	—	— [0PAST]
<i>taka-</i>	<i>-k(u)</i>	+at-	-ta [+PAST]

What remains to be discussed are the relationship between the conclusive form *taka-i* and the attributive form *taka-(k)i*, and the conditions that determine their distribution in relation to the conditional form *taka-ke* and the conjunctive form *taka-ku*. The first two forms, *taka-i* and *taka-(k)i*, share the VA radical *taka-* and a zero tense suffix *-0* specified as [–PAST], but differ in the shape of the VA thematic suffix: *-i* in the conclusive form and *-(k)i* in the attributive form. Correspondingly, they occur in distinct syntactic environments:

(31) *Utukusi-(k)i hito* "A beautiful person":

[_{NP} [_{CP} [_{TP} [_{VAP} *op_i* *utukusi-(k)i*] -0] -0] *hito_i*] → [_{NP} [_{CP} [_{TP} [_{VAP} *op_i* *t_a*] *utukusi-(k)i_a*-0] -0] *hito_i*]¹

¹ There is also the possibility of [_{NP} [_{CP} [_{TP} [_{VAP} *op_i* *utukusi-(k)i*] -0] -0] *hito_i*] → [_{NP} [_{CP} *op_i* [_{TP} [_{VAP} *t_i* *t_a*]]

(32) *Kanozyo ga² utukusi-i* "She is beautiful":

[_{TP} [_{AP} *kanozyo utukusi-i*] -0] → [_{TP} *kanozyo_i-ga* [_{AP} *t_i t_a*] *utukusi-i_a-0*]

As shown in (31), when an VAP functions as the complement of TP headed by T -0 [-PAST] and this TP is embedded within an NP, the VA thematic suffix appears in the form *-(k)i*. By contrast, when an VAP functions as the complement of TP headed likewise by T -0 [-PAST] but the TP is not embedded within an NP, as in (32), the VA thematic suffix appears in the form *-i*. As for the remaining forms *-ke* and *-ku*, *-ke* in modern Japanese is exclusively distributed in the environment preceding the TP-head suffix *-reba* as in (33):

(33) *Kanozyo ga utukusi-ke-reba* "If she is beautiful":

[_{TP} [_{AP} *kanozyo utukusi-ke*] -reba] → [_{TP} *kanozyo_i-ga* [_{AP} *t_i t_a*] *utukusi-ke_a-reba*]

while *-ku* is the form taken by the VA thematic suffix in all other syntactic environments.

5. Structural similarities with verb inflected forms

The following examines the inflected forms of the vowel-stem verb *nobiru* "to stretch/extend" (intransitive) and its classical form *nobu*, and demonstrates the significant structural parallels between these forms and the inflected forms of the verbal adjective *takai* "to be high/tall" and its classical form *takasi*.

The one-argument verb *nobiru* is a non-accusative one, with the stem *nobi-*. On the other hand, the morphologically and semantically corresponding two-argument and accusative verb *noberu* "to stretch/extend" (transitive) has the stem *nobe-*. The shared element *nob-* in both the non-accusative verb stem *nobi-* and the accusative verb stem *nobe-* is present in all the inflected forms of these two verbs. Moreover, since the meaning of "stretching/extension" as a state change is common to all their inflected forms, it can be concluded that *nob-* represents the state change of "stretching/extension". In modern Japanese, although the accusative verb corresponding to *nobiru* is more commonly *nobasu*, the stem *nobas-* of this verb also contains *nob-*, which represents the same state change.

Thus, the stem *nobi-* contains a morpheme boundary within it, as in *nob-i-*. Since neither *nob-* nor *-i* can be further segmented into smaller sequential units, *nob-* is the immediate constituent of the verbal stem *nob-i-* and, denoting the core lexical meaning of the verb, can be considered, alongside the VA radical discussed in Section 3, the radical of the verb — the **V radical**.

utukusi-(k)i_a-0] -0] *hito_i*]. However, this paper employs a simplified model focusing on the allomorphy of the VA thematic suffix *-i* ≈ *-(k)i* ≈ *-ku* ≈ *-ke* and on its syntactic conditions, leaving full syntactic verification to future work.

² *Kanozyo wa utsukusi-i* is a much more natural expression, with the subject *Kanozyo* topicalised through *-wa*: [_{TopicP} *kanozyo_i-wa* [_{TP} *t_i* [_{AP} *t_i t_a*] *utsukusi-i_a-0*] -0].

On the other hand, the *-i* in *nob-i-* combines with the V radical *nob-* to form the stem of the verb — the **V stem** — and it is assumed to function to licence the V radical as a category V at the sentential level. Thus, in parallel with the VA thematic suffix discussed in Section 3, *-i* can be considered the thematic suffix of the verb — the **V thematic suffix**. This V thematic suffix *-i* is also assumed to confer non-accusativity and a one-argument structure as its valency to the V radical *nob-* through its opposition to *-e* in the accusative verb stem *nob-e-*, which, like *-i*, is a V thematic suffix.

Based on the above analysis, the morphological structure of the following inflected forms of VA *takai* — *taka-i-0*, *taka-(k)i-0*, *taka-ku*, *taka-ke-reba*, *taka-k+at-ta* — and their corresponding forms of V *nobiru* — *nob-i-ru*, *nob-i-ru*, *nob-i*, *nob-i-reba*, *nob-i-ta* — is as shown in (34):

(34)

Stem: sentential			Ending
Stem: lexical		Light verb stem	Inflexional suffix
Radical	Thematic suffix		
<i>taka-</i>	<i>-i</i>	—	<i>-0</i> [–PAST]
<i>nob-</i>	<i>-i</i>	—	<i>-ru</i> [–PAST]
<i>taka-</i>	<i>-(k)i</i>	—	<i>-0</i> [–PAST]
<i>nob-</i>	<i>-i</i>	—	<i>-ru</i> [–PAST]
<i>taka-</i>	<i>-ku</i>	—	— [0 PAST]
<i>nob-</i>	<i>-i</i>	—	— [0 PAST]
<i>taka-</i>	<i>-ke</i>	—	<i>-reba</i>
<i>nob-</i>	<i>-i</i>	—	<i>-reba</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+at</i>	<i>-ta</i> [+PAST]
<i>nob-</i>	<i>-i</i>	—	<i>-ta</i> [+PAST]

Except that the tense [–PAST] is encoded by *-0* in VA *taka-i-0* and by *-ru* in V *nob-i-ru*, and that, in VA, the stem *at-* of the light verb *ar-u* may be inserted at the sentence level, the basic structure is the same. The inflected forms of the corresponding classical forms *takasi* and *nobu* exhibit even greater structural similarity, as indicated in (35):

(35)

Stem: sentential			Ending
Stem: lexical		Light verb stem	Inflexional suffix
Radical	Thematic suffix		
<i>taka-</i>	<i>-si</i>	—	<i>-0</i> [–PAST]
<i>nob-</i>	<i>-u</i>	—	<i>-0</i> [–PAST]
<i>taka-</i>	<i>-ki</i>	—	<i>-0</i> [–PAST]
<i>nob-</i>	<i>-u</i>	—	<i>-ru</i> [–PAST]
<i>taka-</i>	<i>-ku</i>	—	— [0PAST]
<i>nob-</i>	<i>-i</i>	—	— [0PAST]
<i>taka-</i>	<i>-ke</i>	—	<i>-reba</i>
<i>nob-</i>	<i>-u</i>	—	<i>-reba</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+ar-i</i>	<i>-ki</i> [+PAST]
<i>nob-</i>	<i>-i</i>	—	<i>-ki</i> [+PAST]

As the VA thematic suffix shows allomorphy $-si \approx -ki \approx -ku \approx -ke$ depending on the syntactic environment, the V thematic suffix also exhibits the alternation $-i \approx -u$ according to the syntactic context in Antient Japanese. Furthermore, the conclusive form *nob-u-0* of the classical *nobu*, corresponding to the modern *nobiru*, like the conclusive form *taka-si-0* of the classical *takasi*, corresponding to the modern *takai*, denoted [–PAST] with *-0*, which is different from modern Japanese.

The only instance where the parallelism in the suffixes breaks down is in the attributive forms *taka-ki-0* and *nob-u-ru*, but there seems to be a reason for this. In VA *takasi*, the opposition between the conclusive form and the attributive form was maintained by the alternation of the thematic suffix allomorphs $-si \approx -ki$, as in *taka-si-0* and *taka-ki-0*, whereas in V *nobu*, the thematic suffix allomorph was the same in both forms — namely, *-u* — and the opposition was instead maintained by the alternation of the past tense suffix allomorphs $-0 \approx -ru$, as in *nob-u-0* and *nob-u-ru*.

6. Analysis summary — Conclusion

(i) The morphological structure of the inflected forms of verbal adjectives is as shown in (36):

(36)

Stem: sentential			Ending
VA stem: lexical		Light verb stem	Inflexional suffix(es)
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-i</i> ³	—	<i>-0</i> [–PAST]
<i>taka-</i>	<i>-(k)i</i>	—	<i>-0</i> [–PAST]
<i>taka-</i>	<i>-ke</i>	—	<i>-reba</i>
<i>taka-</i>	<i>-ku</i>	—	— [0 PAST]
<i>taka-</i>	<i>-ku</i>	—	<i>-te</i> [0 PAST]
<i>taka-</i>	<i>-ku</i>	—	<i>-temo</i>
<i>taka-</i>	<i>-ku</i>	—	<i>-tatte</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-ta</i> [+PAST]
<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-tar-ō</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-tar-a(ba)</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+at-</i>	<i>-tar-i</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+ar-</i>	<i>-ō</i>
<i>taka-</i>	<i>-k(u)</i>	<i>+ar-</i>	<i>-e</i>

(ii) The VA radical is the element that encodes the lexical meaning specific to each verbal adjective. It can appear independently as a constituent of derived or compound words — for

³ Given that *yo-si* and *na-si*, the classical forms of the verbal adjectives *yo-i*, *i-i* "to be good" and *na-i* "(there) not to be" survive in fossilised forms in modern Japanese, it is also possible to assume *-si* as the underlying form, with *s-* being deleted on the surface in all but exceptional cases.

example, *taka-* in *taka-sa* "height", *taka-mi* "loftiness", *taka+nozomi* "high hope" and *taka+dai* "high platform" — but it cannot, on its own, be used as a syntactic constituent belonging to the category VA at the sentential level — for example, **Kare wa se ga taka* "He is tall"; **Asoko ni se no taka ooto ga iru* "There is a tall man over there"; **Koko kara taka sobieru tō ga mieru* "From here, a tower rising high can be seen"; **Se wa taka-reba ii to yū wake de wa nai* "It is not always the case that it is good if the height is tall".

(iii) The function of the VA thematic suffix $-i \approx -(k)i \approx -ku \approx -ke$ is to licence the VA radical as a syntactic constituent of category VA within sentential phrase structure. As noted in (ii), *taka-* on its own cannot be used as a VA in a sentence. Only when it combines with $-i$, $-(k)i$, $-ku$ or $-ke$ — forming *taka-i*, *taka-(k)i*, *taka-ku* or *taka-ke*, for example — does it become syntactically eligible to function as a constituent of category VA — e.g. *Kare wa se ga taka-i*; *Asoko ni se no taka-i ooto ga iru*; *Koko kara taka-ku sobieru tō ga mieru*; **Se wa taka-ke-reba ii to yū wake de wa nai*.

(iv) The form in which the VA thematic suffix appears — $-i$, $-(k)i$, $-ku$ or $-ke$ — is determined by the syntactic environment in which the VA stem occurs, as shown in (37)-(40):

(37) When a VAP headed by a VA stem serves as the complement of a TP headed by the inflexional suffix *-reba*: → ***-ke***

(38) When a VAP headed by a VA stem serves as the complement of a TP headed by -0 [–PAST], and that TP is embedded within an NP: → ***-(k)i***

(39) When a VAP headed by a VA stem serves as the complement of a TP headed by -0 [–PAST], and that TP is not embedded within an NP: → ***-i***

(40) In all other cases: → ***-ku***

(v) At the surface level, $-ku$ exhibits the phonological alternation $[-ku] \approx [-k]$, with vowel deletion occurring only when the VA stem combines with the stem *ar-* of the light verb *ar-u*, resulting in the form $[-k]$. This vowel deletion can be taken to indicate that the VA stem in $-ku$ and the light verb stem *ar-* have come to form a single compound stem at the sentential level.

(vi) As shown in Section 5, there are significant structural parallels between the inflected forms of verbal adjectives — e.g. *takai* and its classical form *takasi* — and those of verbs, especially vowel-stem verbs — e.g. *nobiru* and its classical form *nobu*. Here, the vowel-stem verb *kuru* "to come" is further examined, and the morphological structure of its inflected forms, semantically corresponding to those of the verbal adjective *takai* exemplified in (37), is shown in (41). Again, a strong parallelism in morphological structure can be observed:

(41)

V stem		Ending
V radical	V thematic suffix	Inflexional suffix(es)
<i>k-</i>	<i>-u</i>	<i>-ru</i> [–PAST]
<i>k-</i>	<i>-u</i>	<i>-ru</i> [–PAST]
<i>k-</i>	<i>-u</i>	<i>-reba</i>
<i>k-</i>	<i>-i</i>	— [0 PAST]
<i>k-</i>	<i>-i</i>	<i>-te</i> [0 PAST]
<i>k-</i>	<i>-i</i>	<i>-temo</i>
<i>k-</i>	<i>-i</i>	<i>-tatte</i>
<i>k-</i>	<i>-i</i>	<i>-ta</i> [+PAST]
<i>k-</i>	<i>-i</i>	<i>-tar-ō</i>
<i>k-</i>	<i>-i</i>	<i>-tar-a(ba)</i>
<i>k-</i>	<i>-i</i>	<i>-tar-i</i>
<i>k-</i>	<i>-o</i>	<i>-jō</i>
<i>k-</i>	<i>-o</i>	<i>-i</i>

In vowel-stem verbs, the stem-final vowels — *-u* $\approx$ *-i* $\approx$ *-o* in the inflected forms of the verb *k-u-ru* as well as *-i* in those of the verb *nob-i-ru* in (34) and *-u* $\approx$ *-i* in those of the corresponding classical verb *nob-u-o* in (35) — can be understood as functioning as the thematic suffix.

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